

AdaBoost Learner Weight Formula Derivation

Assume binary classification with labels

$$y_i \in \{-1, +1\}, \quad f_m(x_i) \in \{-1, +1\}.$$

At boosting round m , AdaBoost updates the current ensemble model as

$$F_m(x) = F_{m-1}(x) + \eta_m f_m(x),$$

where:

- $F_{m-1}(x)$ is the current ensemble prediction,
- $f_m(x)$ is the new weak learner,
- η_m is the weight assigned to this learner.

The idea is to choose η_m so that the exponential loss of the updated ensemble is minimized.

Step 1: Write the exponential loss after adding the new learner

$$L_m = \sum_{i=1}^N \exp(-y_i F_m(x_i)).$$

Substituting

$$F_m(x_i) = F_{m-1}(x_i) + \eta_m f_m(x_i),$$

we get

$$L_m = \sum_{i=1}^N \exp\left(-y_i (F_{m-1}(x_i) + \eta_m f_m(x_i))\right).$$

Explanation: This is the total exponential loss after adding the m -th learner.

Step 2: Separate the old ensemble part and the new learner part

Using

$$e^{a+b} = e^a e^b,$$

we write

$$L_m = \sum_{i=1}^N \exp(-y_i F_{m-1}(x_i)) \exp(-\eta_m y_i f_m(x_i)).$$

Now define

$$w_i^{(m)} = \exp(-y_i F_{m-1}(x_i)).$$

Then

$$L_m = \sum_{i=1}^N w_i^{(m)} \exp(-\eta_m y_i f_m(x_i)).$$

Explanation: The quantity $w_i^{(m)}$ acts like the weight of sample i at round m . Samples that are currently hard to classify have larger weights.

Step 3: Use the fact that $y_i f_m(x_i) \in \{-1, +1\}$

There are two cases:

- If sample i is correctly classified, then

$$y_i = f_m(x_i) \Rightarrow y_i f_m(x_i) = 1.$$

So its contribution becomes

$$\exp(-\eta_m).$$

- If sample i is misclassified, then

$$y_i \neq f_m(x_i) \Rightarrow y_i f_m(x_i) = -1.$$

So its contribution becomes

$$\exp(\eta_m).$$

Hence

$$L_m = e^{-\eta_m} \sum_{i: y_i = f_m(x_i)} w_i^{(m)} + e^{\eta_m} \sum_{i: y_i \neq f_m(x_i)} w_i^{(m)}.$$

Explanation: Correctly classified points get multiplied by $e^{-\eta_m}$, while misclassified points get multiplied by e^{η_m} .

Step 4: Express the sums using weighted classification error

Let the weights be normalized so that

$$\sum_{i=1}^N w_i^{(m)} = 1.$$

Define the weighted error of learner f_m as

$$\varepsilon_m = \sum_{i: y_i \neq f_m(x_i)} w_i^{(m)}.$$

Then

$$\sum_{i: y_i = f_m(x_i)} w_i^{(m)} = 1 - \varepsilon_m.$$

So the loss becomes

$$L_m = (1 - \varepsilon_m)e^{-\eta_m} + \varepsilon_m e^{\eta_m}.$$

Explanation: Now the loss is written only in terms of η_m and the weighted error ε_m .

Step 5: Differentiate with respect to η_m

We now minimize

$$L_m(\eta_m) = (1 - \varepsilon_m)e^{-\eta_m} + \varepsilon_m e^{\eta_m}.$$

Differentiate:

$$\frac{dL_m}{d\eta_m} = -(1 - \varepsilon_m)e^{-\eta_m} + \varepsilon_m e^{\eta_m}.$$

Set derivative to zero:

$$-(1 - \varepsilon_m)e^{-\eta_m} + \varepsilon_m e^{\eta_m} = 0.$$

Explanation: We are finding the value of η_m that minimizes the exponential loss.

Step 6: Solve for η_m

Rearranging,

$$\varepsilon_m e^{\eta_m} = (1 - \varepsilon_m)e^{-\eta_m}.$$

Multiply both sides by e^{η_m} :

$$\varepsilon_m e^{2\eta_m} = 1 - \varepsilon_m.$$

So

$$e^{2\eta_m} = \frac{1 - \varepsilon_m}{\varepsilon_m}.$$

Taking logarithm on both sides,

$$2\eta_m = \ln\left(\frac{1 - \varepsilon_m}{\varepsilon_m}\right).$$

Hence

$$\boxed{\eta_m = \frac{1}{2} \ln\left(\frac{1 - \varepsilon_m}{\varepsilon_m}\right)}$$

Explanation: This is the learner weight used in AdaBoost.

Interpretation of the formula

- If ε_m is small, then η_m is large. So a more accurate learner gets a larger vote.
- If $\varepsilon_m = 0.5$, then

$$\eta_m = 0.$$

So a learner no better than random guessing gets no influence.

- If $\varepsilon_m > 0.5$, then $\eta_m < 0$. Such a learner is worse than random and should not be used as-is.

Derivation of the Sample Weight update rule

Recall that at round m , the sample weights were defined as

$$w_i^{(m)} = \exp(-y_i F_{m-1}(x_i)).$$

After adding the new learner, the ensemble becomes

$$F_m(x) = F_{m-1}(x) + \eta_m f_m(x).$$

To preserve the same interpretation at the next round, define the new weights as

$$w_i^{(m+1)} = \exp(-y_i F_m(x_i)).$$

Substituting the updated ensemble expression,

$$w_i^{(m+1)} = \exp(-y_i (F_{m-1}(x_i) + \eta_m f_m(x_i))).$$

Using

$$e^{a+b} = e^a e^b,$$

we obtain

$$w_i^{(m+1)} = \exp(-y_i F_{m-1}(x_i)) \exp(-\eta_m y_i f_m(x_i)).$$

Since

$$\exp(-y_i F_{m-1}(x_i)) = w_i^{(m)},$$

this becomes

$$\boxed{w_i^{(m+1)} = w_i^{(m)} \exp(-\eta_m y_i f_m(x_i))}$$

Explanation: The sample weights at the next round are obtained directly from the updated exponential loss contribution of each sample.

Interpretation of the update

Since

$$y_i f_m(x_i) = \begin{cases} +1, & \text{if sample } i \text{ is correctly classified} \\ -1, & \text{if sample } i \text{ is misclassified} \end{cases}$$

we get:

– If correctly classified:

$$w_i^{(m+1)} = w_i^{(m)} e^{-\eta_m}$$

– If misclassified:

$$w_i^{(m+1)} = w_i^{(m)} e^{\eta_m}$$

Thus:

- correctly classified samples receive smaller weight,
- misclassified samples receive larger weight.

Why this update is consistent

At every boosting round,

$$w_i^{(m)} = \exp(-y_i F_{m-1}(x_i))$$

remains true.

Hence, the sample weights always represent the exponential-loss contribution of each training point under the current ensemble.

This ensures that samples contributing more to the loss automatically receive more attention in the next round.